Problem 1

Heun method (2nd order Runge-Kutta)

$$2\frac{d^2y}{dx^2} + 3x^2y\frac{dy}{dx} = 10x^2$$

@ $x=1$, $y=2$, $\frac{dy}{dx}=2$

$$\frac{dy}{dx} = f(y,x)\ ;$$

$$\therefore 2\frac{df}{dx} + 3x^2yf = 10x^2$$

$$\frac{df}{dx} = 5x^2 - \frac{3}{2}x^2yf = g(x,y)$$

@ $x_0=1$, $f(1,2)=2$, $y=2$, $g(1,2)=-1$

Discretize the domain : $x^n = x_0 + n\Delta x$ (1, 0.1)

Discretize the ODE : $\frac{\tilde{y}^{n+1} - y^n}{\Delta x} = f(y^n, x^n) \rightarrow \tilde{y}^{n+1} = y^n + \Delta x\, f(y^n, x^n)$

$$\frac{f^{n+1} - f^n}{\Delta x} = \frac{f(y^n,x^n) + g(\tilde{y}^{n+1}, x^{n+1})}{2}$$

$$f^{n+1} = f^n + \frac{\Delta x}{2} g(\tilde{y}^{n+1}, x^{n+1})$$

$$f(y^{n+1}, x^{n+1}) = f(y^n, x^n) + \frac{\Delta x}{2}\left[5(x^{n+1})^2 - \frac{3}{2}(x^{n+1})^2 \tilde{y}^{n+1} f(\tilde{y}^{n+1}, x^{n+1})\right]$$

$f^{n+1} = f(y^{n+1}, x^{n+1})$; $f(x_0, y_0) = 2$; $g(x_0, y_0) = -1$

n	$x^n = x_0 + n\Delta x$	$\tilde{y}^n = y^{n-1} + \Delta x f^{n-1}$	$\tilde{f}^n = f^{n-1} + \Delta x\, g(\tilde{y}^n, x^n)$	$f^n = f^{n-1} + \frac{\Delta x}{2} g(\tilde{y}^n, x^n)$	$y^n = y^{n-1} + \frac{\Delta x}{2}(f^{n-1} + \tilde{f}^n)$
0	$x^0 = x_0 = 1$	—	—	—	—
1	$x^1 = 1 + (0.1) = 1.1$	$\tilde{y}^1 = y^0 + \Delta x f^0 = 2 + (0.1)(2) = 2.2$	$\tilde{f}^1 = f^0 + \Delta x\, g(\tilde{y}^1, x^1) = 2 + (0.1)(-1) = 1.9$	$f^1 = f^0 + \frac{\Delta x}{2} g(\tilde{y}^1, x^1) = 2 + \frac{(0.1)}{2}\left(5(1.1)^2 - \frac{3(1.1)^2(2.2)}{2}(1.9)\right) = 1.8732$	$y^1 = y^0 + \frac{(0.1)}{2}(2 + 1.9) = 2.195$
2	$x^2 = 1 + 2(0.1) = 1.2$	$\tilde{y}^2 = y^1 + \Delta x f^1 = 2.195 + (0.1)(1.8732) = 2.38823$	$\tilde{f}^2 = f^1 + \Delta x\, g(\tilde{y}^2, x^2) = 1.8732 + (0.1)(-1.4183) = 1.7732$	$f^2 = f^1 + \frac{\Delta x}{2} g(\tilde{y}^2, x^2) = 1.8732 + \frac{(0.1)}{2}\left(5(1.2)^2 - \frac{3(1.2)^2}{2}(2.38823)(1.732)\right) = 1.7072$	$y^2 = y^1 + \frac{(0.1)}{2}(f^1 + \tilde{f}^2) = 2.195 + \frac{(0.1)}{2}(1.8732 + 1.732) = 2.3$

$5(1.1)^2 - \frac{3}{2}(1.1)^2(2.195)(1.8732) = -1.4183$

Therefore ;

for $\Delta x = 0.1$ (w/ the given IC's)

we get

$y^1 = 2.195$, $f^1 = \frac{dy^1}{dx} = 1.8732$	@ $x = 1.1$
$y^2 = 2.3952$, $f^2 = \frac{dy^2}{dx} = 1.7172$	@ $x = 1.2$

Problem 2

2nd order Adams-Moulton method

$$2\frac{dy}{dx} - 3y^2 = 10 \quad , \quad y=10 \text{ at } x=1$$

$$\frac{dy}{dx} = \frac{5 - 3y^2}{2} = f(y,x) \ ; \quad y_0(x_0=1) = 10$$

$$f(y_0, x_0) = \frac{5 - 3(10)^2}{2} = -145$$

Implicit Adams-Moulton (Trapezoidal Method) :

$$y_{n+1} = y_n + \frac{\Delta x}{2}\left[f(y_n, x_n) + f(y_{n+1}, x_{n+1})\right]$$

@ $n=0$, $y_1 = y_0 + \frac{\Delta x}{2}\left[f(y_0,x_0) + f(y_1,x_1)\right]$

$$y_1 = 10 + \frac{(0.1)}{2}\left[-145 + 5 - 3y_1^2/2\right]$$

$$y_1 = 10 - 7 - 0.075\, y_1^2$$

$$0.075\, y_1^2 + y_1 - 3 = 0$$

Solving using Newton Rhapson : Let $h = 0.075\, y_1^2 + y_1 - 3$

$$h' = 0.15\, y_1 + 1$$

$$y_1^{(k+1)} = y_1^{(k)} - \frac{h(y_1^{(k)})}{h'(y_1^{(k)})}$$

@ $k=0$,

$$y_1^{(1)} = y_1^{(0)} - \frac{h(y_1^{(0)})}{h'(y_1^{(0)})}$$

estimate using Euler → $y_1^{(0)} = y_0 + \Delta x\, f(y_0) = 10 + (0.1)(-145) = -4.5$

$h(y_1^{(0)}) = h(-4.5) = 0.075(-4.5)^2 + (-4.5) = -5.921875$

$h'(-4.5) = 0.15(-4.5) + 1 = 0.325$

2 BVP → Problem 3 → L & T

$\therefore\ y_1^{(1)} = 13.9$

@ k=1,

$h(13.9) = 0.075(13.9)^2 + 13.9 - 3 = 25.39075$

$h'(13.9) = 0.15(13.9) + 1 = 3.085$

$y_1^{(2)} = 13 - \frac{25.39075}{3.085} = 5.67 \rightarrow y_1 \sim 5.67$

Using quadratic formula

$y_1 = \{-15.86,\ 2.5\}$ ✓ → still far off w/ two iterations but will just use this value

@ n=1 ; $y_2 = y_1 + \frac{\Delta x}{2}\left[f(y_1, x_1) + f(y_2, x_2)\right]$

$f(y_1, x_1) = 5 - \frac{3}{2}y_1^2 = 5 - \frac{3}{2}(5.67)^2 = -43.22$

$y_2 = y_1 + \frac{\Delta x}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$ (5.67)

$y_2 = 5.67 + \frac{(0.1)}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$

$y = 3.759 + 0.075y_2^2$

$0.075y_2^2 + y - 3.759 = 0$

Solving using Newton Rhapson : Let $g = 0.075y_2^2 + y_2 - 3.759$

$g' = 0.15y_2 + 1$

$y_2^{(k+1)} = y_2^{(k)} - \frac{g(y_2^{(k)})}{g'(y_2^{(k)})}$

@ k=0,

$y_2^{(1)} = y_2^{(0)} - \frac{g(y_2^{(0)})}{g'(y_2^{(0)})}$

estimate using Euler → $y_2^{(0)} = y_1 + \Delta x f(y_1) = 5.67 + (0.1)(-43.22) = 1.348$

$g(y_2^{(0)}) = g(1.348) = 0.075(1.348)^2 + (-2.411) = -2.2747$

$g'(1.348) = 0.15(1.348) + 1 = 1.2022$

$\therefore\ y_2^{(1)} = 1.348 - \frac{(-2.2747)}{(1.2022)} = 3.24$

@ k=1,

$g(13.9) = 0.075(3.24)^2 + 3.24 - 3.759 = 0.26832$

$g'(13.9) = 0.15(3.24) + 1 = 1.486$

$y_2^{(2)} = 3.24 - \frac{0.26832}{1.486} = 3.0594 \rightarrow y_2 \sim 3.06$

Using quadratic formula

$y_2 = \{-16.39,\ 3.0587\}$ ✓ → much better approximation

$\therefore$ For

$x_1 = 1.1,$	$y_1 = 5.67$	w/ $\Delta x = 0.1$
$x_2 = 1.2,$	$y_2 = 3.06$	& given initial conditions

Problem 3

Steady state conduction in a flat plate

L ⇒ plate thickness

$\dot{q}$ ⇒ heat generation

$\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q} = 0$

$k = a + bT$

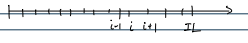
$\dot{q} = d + eT$

$-k\frac{dT}{dx}\Big|_{x=0} = h_H\left(T_H - T(x=0)\right)$

$-k\frac{dT}{dx}\Big|_{x=L} = h_L\left(T(x=L) - T_c\right)$

$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$ } $F_1(T_1, T_2)$ ($k_1 = a + bT_1$)

$-k_1\frac{(T_{IL} - T_I)}{\Delta x} = h_L(T_{IL} - T_c)$ } $F_{IL}(T_{IL}, \ldots)$



Discretizing the equation :

$\left[\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q}\right]_i = \frac{k\frac{dT}{dx}\Big|_{i+\frac{1}{2}} - k\frac{dT}{dx}\Big|_{i-\frac{1}{2}}}{\Delta x} + q_i = 0$

$= \frac{\left(k_{i+\frac{1}{2}}\frac{T_{i+1} - T_i}{\Delta x}\right) - \left(k_{i-\frac{1}{2}}\right)\frac{T_i - T_{i-1}}{\Delta x}}{\Delta x} + d + eT_i = 0$ } $F_i(T_{i+1}, T_i, T_{i-1}) = 0$, $i = 2, 3, \ldots, IL-1$

$k_{i+\frac{1}{2}} = \frac{k_i + k_{i+1}}{2}$

$$= \frac{k_{i+\frac{1}{2}}\left(\frac{\Delta x}{2}\right)^2 + k_{i-\frac{1}{2}}\left(\frac{\Delta x}{2}\right)^2}{\Delta x} + d + eT_i = 0$$

$$k_{i+\frac{1}{2}} = \frac{k_i + k_{i+1}}{2}$$

$$k_{i-\frac{1}{2}} = \frac{k_i + k_{i-1}}{2} = \frac{(a+bT_i) + (a+bT_{i-1})}{2}$$

$$F_i(T_{i+1}, T_i, T_{i-1}) = 0, \quad i = 2, 3, \ldots, IL-1$$

Translating this in the form of $Ax = b$:

$$\frac{(k_{i-1/2})}{(\Delta x)^2} T_{i-1} - \left[\frac{(k_{i+1/2})}{(\Delta x)^2} + \frac{(k_{i-1/2})}{(\Delta x)^2} - e\right] T_i + \frac{(k_{i-1/2})}{(\Delta x)^2} T_{i-1} = -d$$

Need the Newton Raphson (product of T_{ij}) → find derivatives

@ boundaries:

F_1 & F_{IL} apply

∴ Matrix A will be constructed as:

$$\begin{bmatrix} A_{i,i+1} = \frac{k_{i+1/2}}{(\Delta x)^2} & & \\ & A_{i,i} = -\left(\frac{(k_{i+1/2}) + (k_{i-1/2})}{(\Delta x)^2} + e\right) & \\ & & A_{i,i-1} = \frac{(k_{i-1/2})}{(\Delta x)^2} \end{bmatrix} \begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} BC\#1 \\ -q_i \\ BC\#2 \end{bmatrix}$$

Boundary conditions

$$-k\left.\frac{dT}{dx}\right|_{x=0} = h_H(T_H - T(x=0))$$

$$-k\left.\frac{dT}{dx}\right|_{x=L} = h_L(T(x=L) - T_c)$$

$$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$$

$$-k_1\frac{(T_{IL} - T_{IL-1})}{\Delta x} = h_L(T_{IL} - T_c)$$

$$\frac{-k_1}{\Delta x}T_2 + \frac{k_1}{\Delta x}T_1 = h_H T_H - h_H T_1$$

$$\left(\frac{k_1}{\Delta x} + h_H\right)T_1 - \frac{k_1}{\Delta x}T_2 = h_H T_H$$

$A[0,0]$ $A[0,1]$ boundary $[0]$

$$\frac{-k_1}{\Delta x}T_{IL-1} + \frac{k_1}{\Delta x}T_{IL} = h_L(T_{IL} - T_c)$$

$$\left(\frac{-k_1}{\Delta x} - h_L\right)T_{IL} + \left(\frac{k_1}{\Delta x}\right)T_{IL-1} = -h_L T_c$$

$A[-1,-1]$ $A[-1,-2]$

$$F(T_{i-1}, T_i, T_{i+1}) = (a+bT$$

Newton Raphson

$$\begin{bmatrix} \frac{\partial F_1}{\partial T_1} & \frac{\partial F_2}{\partial T_2} \\ \cdots & \cdots \end{bmatrix}$$

The above algorithm is coded in python (w/ the help of Chatgpt to debug the code) and the following plots are obtained

The decrease temperature (from hot end of ~2000 K) to cold end of ~ 400 K makes sense since the rod is cooling down.

Additionally, the grid spaces were varied from 100 to 10000 to give the same results (grid independent)

Thomas

Office hrs:

$u(x,y) = 0$ } Need to solve for this → start w/ a guess

$v(x,y) = 0$

$$\begin{bmatrix} \frac{\partial u}{\partial x}\Big|_{(x_i, y_i)} & \frac{\partial u}{\partial y}\Big|_{(x_i, y_i)} \\ \frac{\partial v}{\partial x}\Big|_{x_i, y_i} & \frac{\partial v}{\partial y}\Big|_{x_i, y_i} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = - \begin{bmatrix} u(x_i, y_i) \\ v(x_i, y_i) \end{bmatrix}$$

Constants

Constants

Thomas

$Ax = b$ (Thomas algo.)

↳ Update guess next

$$x^{i+1} = x^i + \Delta x$$
$$y^{i+1}$$

↳ Iterate until

$$\left| (\text{norm}\left[\Delta x^{i+1}, y^{i+1}\right]) - \text{norm}\left[\Delta x^i, y^i\right] \right| < \varepsilon = 0$$

↳ Incorporate the boundary conditions